Homework 08

The former kernel-tuning question has moved to Homework 09.

Solutions for Questions 1 and 2 will be available after Monday, October 19, 2026. No solution can be provided for Question 3.

For Questions 1 and 2, use either R or Python.

Question 1: Boundary and margin

Use `margin-geometry-20.csv` from the combined ZIP, which contains 20 observations with predictors x_1, x_2 and class labels $y \in \{-1, 1\}$. Consider the score

$$f(x) = 0.8x_1 + 0.6x_2 .$$

Since $\| (0.8, 0.6) \|_2 = 1$, the lines $f(x) = \pm 1$ are each a distance 1 from the decision boundary $f(x) = 0$.

a. Calculate the distance from every observation to the decision boundary and its functional margin $yf(x)$. In a table, report these values and whether each observation is inside, on, or beyond the margin for its class. Compare $yf(x)$ with 1 to make this distinction.

b. Plot the observations, the boundary $f(x) = 0$, and the margin lines $f(x) = \pm 1$. Distinguish the two classes and identify the observations inside the margin in your plot.

c. Report how many observations are inside, on, and beyond the margin. Which observations are misclassified by this decision rule?

Use the observation numbers supplied in the data. Set the x_1 axis to $[-3.2, 3.2]$, set the x_2 axis to $[-3, 3]$, and use an equal aspect ratio. Do not fit a new classifier for this question.

ANSWER:

```r
margin20 <- read.csv("data/margin-geometry-20.csv")

margin20
```

```
   observation    x1    x2  y
1            1  0.68  0.76  1
2            2  1.80 -0.15  1
3            3  1.38  0.66  1
4            4  0.88  1.66  1
5            5  0.62  2.34  1
6            6  2.78 -0.04  1
7            7  1.86  1.52  1
8            8  0.50  0.40  1
9            9  0.10  0.20  1
10          10 -0.40  0.20  1
11          11 -0.92 -0.44 -1
12          12 -0.50 -1.50 -1
13          13 -1.90 -0.05 -1
14          14 -0.44 -2.33 -1
```

```
15          15 -1.98 -0.61 -1
16          16 -2.74  0.07 -1
17          17 -1.90 -1.55 -1
18          18 -0.50 -0.40 -1
19          19  0.00 -0.40 -1
20          20  0.40 -0.20 -1
```

```r
names(margin20)
```

```
[1] "observation" "x1"          "x2"          "y"
```

```r
margin20$f <- 0.8 * margin20$x1 + 0.6 * margin20$x2

margin20$distance <- abs(margin20$f)

margin20$functional_margin <- margin20$y * margin20$f
```

```r
margin20$margin_position <- ifelse(
  abs(margin20$functional_margin - 1) < 1e-10,
  "On",
  ifelse(
    margin20$functional_margin < 1,
    "Inside",
    "Beyond"
  )
)
```

```r
margin20$predicted <- ifelse(margin20$f >= 0, 1, -1)

margin20$misclassified <- margin20$predicted != margin20$y
```

```r
result_table <- margin20[, c(
  "observation",
  "x1",
  "x2",
  "y",
  "distance",
  "functional_margin",
  "margin_position"
)]

result_table
```

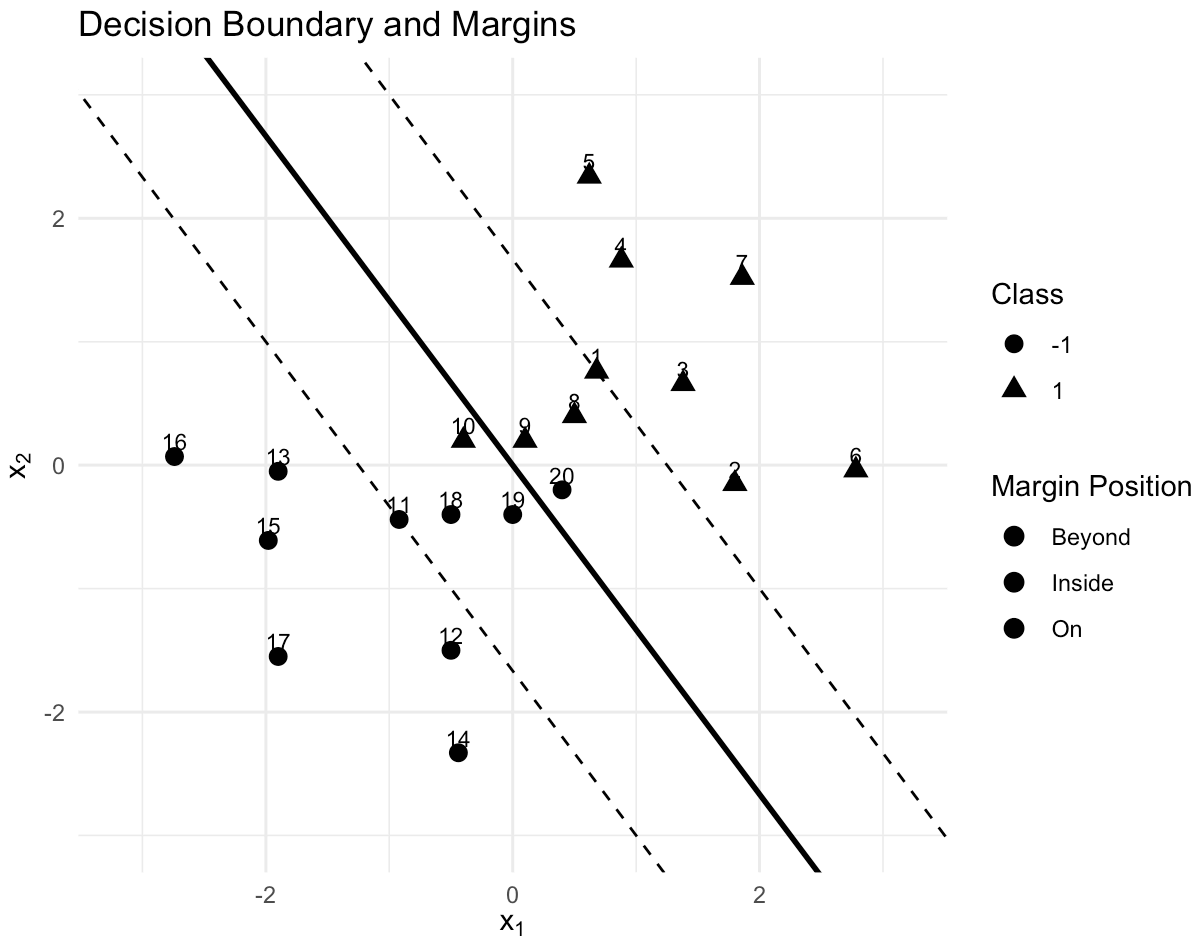
```
   observation    x1    x2  y distance functional_margin margin_position
1            1  0.68  0.76  1     1.00              1.00              On
2            2  1.80 -0.15  1     1.35              1.35          Beyond
3            3  1.38  0.66  1     1.50              1.50          Beyond
4            4  0.88  1.66  1     1.70              1.70          Beyond
5            5  0.62  2.34  1     1.90              1.90          Beyond
6            6  2.78 -0.04  1     2.20              2.20          Beyond
7            7  1.86  1.52  1     2.40              2.40          Beyond
8            8  0.50  0.40  1     0.64              0.64          Inside
```

9	9	0.10	0.20	1	0.20	0.20	Inside
10	10	-0.40	0.20	1	0.20	-0.20	Inside
11	11	-0.92	-0.44	-1	1.00	1.00	On
12	12	-0.50	-1.50	-1	1.30	1.30	Beyond
13	13	-1.90	-0.05	-1	1.55	1.55	Beyond
14	14	-0.44	-2.33	-1	1.75	1.75	Beyond
15	15	-1.98	-0.61	-1	1.95	1.95	Beyond
16	16	-2.74	0.07	-1	2.15	2.15	Beyond
17	17	-1.90	-1.55	-1	2.45	2.45	Beyond
18	18	-0.50	-0.40	-1	0.64	0.64	Inside
19	19	0.00	-0.40	-1	0.24	0.24	Inside
20	20	0.40	-0.20	-1	0.20	-0.20	Inside

```r
library(ggplot2)

ggplot(margin20, aes(x = x1, y = x2)) +
  geom_abline(
    intercept = 0,
    slope = -0.8 / 0.6,
    linewidth = 1
  ) +
  geom_abline(
    intercept = 1 / 0.6,
    slope = -0.8 / 0.6,
    linetype = "dashed"
  ) +
  geom_abline(
    intercept = -1 / 0.6,
    slope = -0.8 / 0.6,
    linetype = "dashed"
  ) +
  geom_point(
    aes(
      shape = factor(y),
      fill = margin_position
    ),
    size = 3
  ) +
  geom_text(
    aes(label = observation),
    nudge_y = 0.12,
    size = 3
  ) +
  coord_fixed(
    xlim = c(-3.2, 3.2),
    ylim = c(-3, 3)
  ) +
  labs(
    x = expression(x[1]),
    y = expression(x[2]),
    shape = "Class",
    fill = "Margin Position",
    title = "Decision Boundary and Margins"
```

```
  ) +
  theme_minimal()
```




```
table(margin20$margin_position)
```

```
Beyond Inside     On
    12      6      2
```

```
margin20[margin20$misclassified, c(
  "observation",
  "y",
  "f",
  "functional_margin"
)]
```

```
   observation  y    f functional_margin
10          10  1 -0.2              -0.2
20          20 -1  0.2              -0.2
```

For each observation, the distance to the decision boundary was calculated using |f(x)|, since the coefficient vector has a norm of 1. The functional margin was calculated as yf(x). Observations with yf(x)<1 were classified as inside the margin, observations with yf(x)=1 were classified as on the margin, and observations with yf(x)>1 were classified as beyond the margin. The calculated distances, functional margins, and classifications are shown in the table above.

The solid line represents the decision boundary f(x)=0, while the two dashed lines represent the margin boundaries f(x)=1 and f(x)=-1. The observations are distinguished according to their true class labels, and the observations that fall inside the margin are identified separately on the plot. Points farther from the decision boundary than their corresponding margin line are beyond the margin, while points between the two margin lines or on the incorrect side of the boundary have functional margins less than 1.

There are 6 observations inside the margin, 2 observations on the margin, and 12 observations beyond the margin. Observations 10 and 20 are misclassified because their functional margins are negative.

Question 2: Margin in four dimensions

Use `margin-geometry-4d.csv` from the combined ZIP, which contains 50 observations with four predictors and class labels $y \in \{-1, 1\}$. Consider the linear score

$$f(\boldsymbol{x}) = \beta_0 + \boldsymbol{x}^{\mathsf{T}} \boldsymbol{\beta}, \qquad \beta_0 = -1, \quad \boldsymbol{\beta} = (1, 2, -2, 4)^{\mathsf{T}}.$$

a. Project the observations onto the direction of $\boldsymbol{\beta}$ and plot their positions on one axis. Distinguish the two classes and mark the decision boundary. Use the plot to calculate the geometric margin M, the distance from the boundary to the closest observation.

b. Express the same decision boundary using the SVM normalization in which the smallest functional margin $y_i f(\boldsymbol{x}_i)$ is set to 1. What are the corresponding values of $\boldsymbol{\beta}$ and β_0?

ANSWER:

```
margin4d <- read.csv("data/margin-geometry-4d.csv")

margin4d
```

```
   observation     x1    x2    x3    x4  y
1            1   8.72 -1.31  1.31 -2.62 -1
2            2   1.94  0.13 -2.38  1.01  1
3            3   1.12 -2.26  0.01 -1.52 -1
4            4  -1.26 -0.27 -1.98  0.96  1
5            5  -3.28 -2.06 -0.94 -0.37 -1
6            6  -6.22 -0.44  0.44  0.62 -1
7            7 -12.26  0.98 -0.98  0.46 -1
8            8   4.26 -1.98 -0.27 -1.71 -1
9            9   2.66 -0.68  2.18 -1.36 -1
10          10   4.84 -0.07 -2.18 -0.14  1
11          11  -6.32  0.11 -0.11 -0.53 -1
12          12   0.32  0.64  0.86  2.03  1
13          13  12.50 -0.50  0.50  0.50  1
14          14   1.82  0.64  0.11  1.28  1
15          15  -6.16  1.18  0.32  0.11 -1
16          16  -4.12 -0.74 -1.51  3.02  1
17          17   2.80 -1.15 -0.35 -1.55 -1
18          18   4.98  1.71  0.54  1.17  1
19          19   6.48 -0.54 -2.46  0.42  1
20          20  -7.80  0.90  1.35  0.30 -1
21          21   1.26 -1.23  0.48 -0.96 -1
22          22  -3.32 -2.14  0.64  0.22 -1
```

```
23          23  -4.90 -1.55 -0.70 -0.85 -1
24          24   7.94 -0.62 -0.13  1.01  1
25          25  -1.86 -2.22  1.47  0.06 -1
26          26   2.56 -2.38  0.88 -1.76 -1
27          27  -0.26 -2.02 -0.98 -1.04 -1
28          28  -2.64  1.47 -1.47  1.44  1
29          29  -7.84  0.07 -0.82 -0.61 -1
30          30  -2.62  0.76  0.74  3.02  1
31          31   0.28  0.56 -0.56  1.12  1
32          32 -10.04  1.67 -2.42  3.34  1
33          33   7.74  1.23  1.02 -0.54  1
34          34  -6.30 -2.10 -0.90  0.30 -1
35          35   4.86  0.72 -1.47 -0.06  1
36          36   2.84  0.43 -0.43 -2.14 -1
37          37   3.52  1.79 -0.29  1.33  1
38          38   7.96  0.17  0.58  1.09  1
39          39   6.30  2.10  0.90 -0.30  1
40          40  -0.36  0.03  1.47 -1.44 -1
41          41   0.30  2.10 -2.10 -0.30  1
42          42   1.92  1.59  0.66  1.68  1
43          43  -8.60  1.55 -1.55  3.10  1
44          44   4.28 -1.19  0.44 -1.63 -1
45          45  -4.90 -0.80  0.05 -0.85 -1
46          46  -1.88 -0.01  2.26 -0.77 -1
47          47   6.40 -0.70 -0.80  0.85  1
48          48   4.92  2.34 -0.09  0.18  1
49          49  -0.42 -2.34  0.84 -0.93 -1
50          50  -1.16 -0.07 -0.68  2.11  1
```

```r
names(margin4d)
```

```
[1] "observation" "x1"          "x2"          "x3"          "x4"
[6] "y"
```

```r
beta <- c(1, 2, -2, 4)
beta0 <- -1

beta_norm <- sqrt(sum(beta^2))
beta_norm
```

```
[1] 5
```

```r
X <- as.matrix(margin4d[, c("x1", "x2", "x3", "x4")])

margin4d$projection <- as.numeric(X %*% beta) / beta_norm
```

```r
boundary_projection <- -beta0 / beta_norm
boundary_projection
```

```
[1] 0.2
```

```r
margin4d$distance <- abs(
  margin4d$projection - boundary_projection
)
```

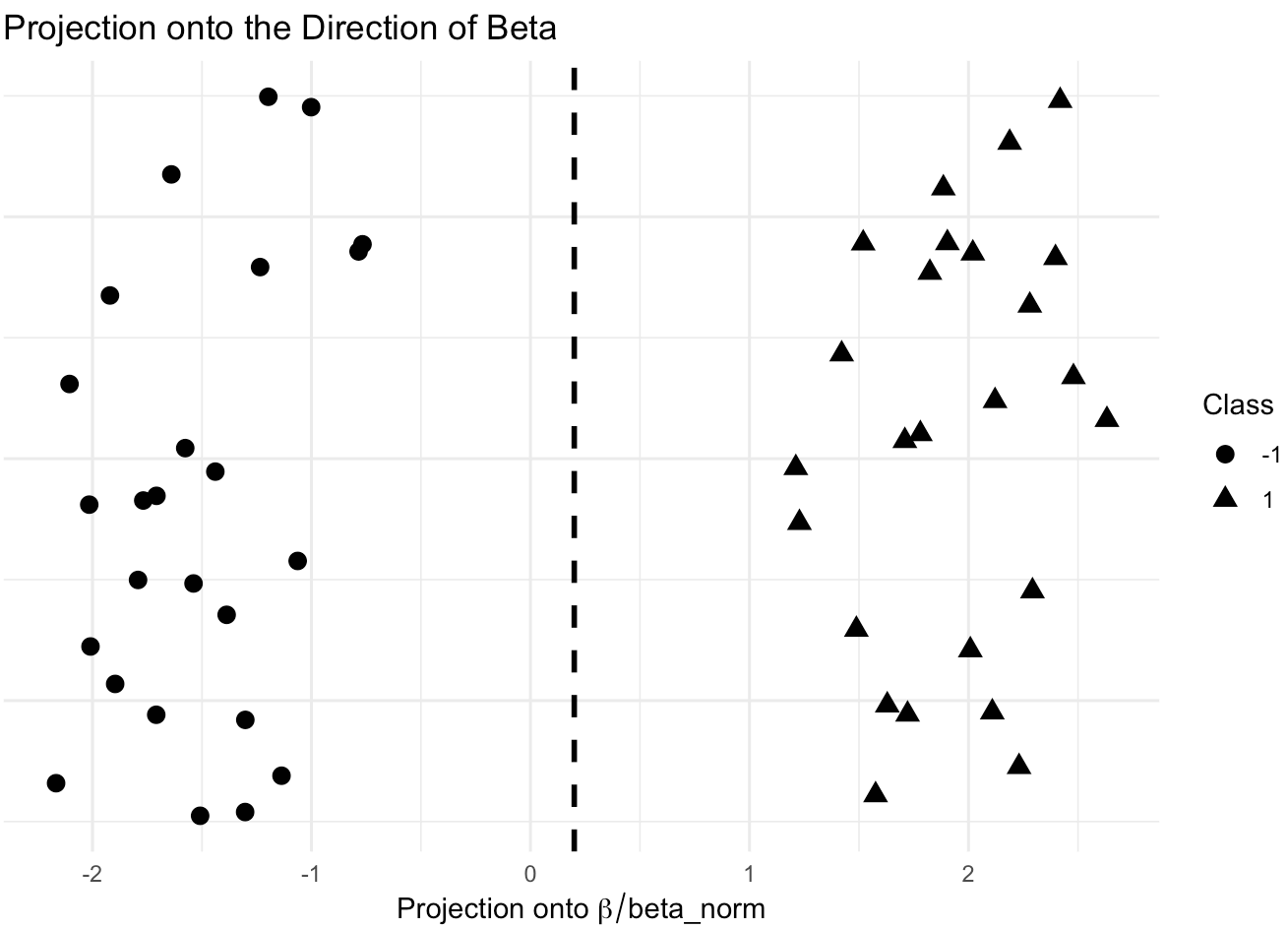
```r
M <- min(margin4d$distance)
M
```

```
[1] 1
```

```r
margin4d[
  abs(margin4d$distance - M) < 1e-10,
]
```

```
   observation    x1    x2    x3    x4  y projection distance
4            4 -1.26 -0.27 -1.98  0.96  1        1.2        1
15          15 -6.16  1.18  0.32  0.11 -1       -0.8        1
33          33  7.74  1.23  1.02 -0.54  1        1.2        1
36          36  2.84  0.43 -0.43 -2.14 -1       -0.8        1
```

```r
ggplot(
  margin4d,
  aes(
    x = projection,
    y = 0,
    shape = factor(y)
  )
) +
  geom_point(size = 3, position = position_jitter(height = 0.03)) +
  geom_vline(
    xintercept = boundary_projection,
    linetype = "dashed",
    linewidth = 1
  ) +
  labs(
    x = expression(paste("Projection onto ", beta / beta_norm)),
    y = NULL,
    shape = "Class",
    title = "Projection onto the Direction of Beta"
  ) +
  theme_minimal() +
  theme(
    axis.text.y = element_blank(),
    axis.ticks.y = element_blank()
  )
```




```
margin4d$f <- beta0 + as.numeric(X %*% beta)

margin4d$functional_margin <- margin4d$y * margin4d$f

min_functional_margin <- min(margin4d$functional_margin)

min_functional_margin
```

```
[1] 5
```

```
beta_svm <- beta / min_functional_margin
beta0_svm <- beta0 / min_functional_margin

beta_svm
```

```
[1]  0.2  0.4 -0.4  0.8
```

```
beta0_svm
```

```
[1] -0.2
```

```
f_svm <- beta0_svm + as.numeric(X %*% beta_svm)

min(margin4d$y * f_svm)
```

```
[1] 1
```

The norm of the coefficient vector is

$$\| \boldsymbol{\beta} \| = \sqrt{1^2 + 2^2 + (-2)^2 + 4^2} = 5.$$

After projecting the observations onto the direction of beta, the decision boundary is located at 0.2 on the projected axis.

The projection plot shows the positions of the two classes relative to the decision boundary and makes the minimum distance between an observation and the boundary visible.

Rescaling does not change the location of the decision boundary because every coefficient, including the intercept, is multiplied by the same positive constant. It only changes the scale of the score so that the closest observation has a functional margin of 1.

The corresponding values of $\boldsymbol{\beta}$ and β_0 are 0.2 0.4 -0.4 0.8 and -0.2.

Question 3: Peer review of a Homework 7 skill

The instructor's GitHub account, **rqzhu-aide**, will open a **GitHub Issue** in the repository containing your Homework 7 submission. It will link to **one** assigned student's repository.

1. **Use the skill.** Find the complete skill definition in the assigned student's Homework 7 report. Copy it without changes into a local `SKILL.md` file in your own folder. You do not need to install it. Give your agent the path to the file and explicitly ask it to follow that skill while analyzing the `SAheart` data, using the response and predictor coding from Homework 7. Provide the necessary data information, but no additional preferences about methods, figures, model selection, or report style. Let the skill direct the analysis.

Monitor the time, your token use, and your weekly quota while the agent works. If the run is taking too long or using your allowance too quickly, you may send a **steer message** asking the agent to finish immediately, or stop the run and summarize what you observed. A steer message redirects an agent while it is working; learn how to send one in your AI tool before starting.

2. **Write a peer review.** Read the student's submitted report and what the skill produced in your run. Give a brief overall assessment, followed by **three major items the author could improve**. You may comment on any aspect of the skill or report. For each item, explain what you observed, why it matters, and what you recommend changing. If you stopped the run early, explain where and why. Write the review yourself, without AI assistance.

3. **Share your review.** Open a GitHub Issue in the assigned student's Homework 7 repository containing your review. Copy the same review into your Homework 8 submission report and include a link to the GitHub Issue.

ANSWER: Overall, the student's submitted skill and report are concise and readable. It included the one-page analysis, a comparison table, a figure, and a separate hold-out test result. However, there is still room for improvement

1. The skill could offer more flexibility when tuning the various models. The submitted report only considers 1 PCA dimension and 1 LDA–QDA blending weight. More blending weights and dimensions could prove to be useful. Performance may improve if we try more possibilities. The skill does not directly specify which values to compare, just to select parameters using cross-validation. So, to prevent limiting our analysis, the skill

should instruct the AI agent to attempt and evaluate multiple different PCA dimensions and blending weights using cross-validation.

2. The skill could offer a bit more guidance for handling QDA estimation problems. The skill utilizes ordinary QDA at a minimum, but it does not offer guidance or direction for when QDA cannot estimate an invertible covariance matrix. Since QDA involves estimating a separate covariance matrix for each class, too many predictors compared to the number of observations can make the estimates singular and unstable. Then, the analysis fails. So, the skill could be improved by checking for the covariance matrix's stability and using some backup plan for when it fails. The skill should clearly report if ordinary QDA cannot be fitted.

3. The skill could offer a more detailed interpretation of its classification errors. The student's report provided a test misclassification rate of 26.6% and a confusion matrix showing 23 false negatives and 14 false positives. However, the report did not explain these results in detail. The analysis did not provide context or clarity. In the context of the SAheart dataset, a false negative means predicting no coronary heart disease for someone whose observed outcome is CHD. This language and level of detail should be explicitly used in the final report page. So, the skill should include a lengthier interpretation of its results in reference to the specific dataset. Additionally, it could discuss measures like sensitivity and specificity in depth.

Link to GitHub Issue: https://github.com/manushrip06/stat432-fall2026/issues/4